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Enhancing Industrial Operator Training through BIM-Enriched Virtual Reality Scenes

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Abstract: In today's evolving industrial landscape, enhancing workforce resilience by improving the ability to navigate complex and unforeseen scenarios, is essential. This paper explores the role of training, especially during disruptions, introducing an innovative approach that combines Building Information Modeling (BIM) and Virtual Reality (VR). This aims to ease the creation of automatic BIM-enriched VR training environments. However, this process revealed some technical challenges that need to be addressed for an effective implementation. Therefore, this paper covers the theoretical background of VR and BIM, their common applications, and develops the conceptual basis of BIM-enriched VR models. A prototype of this integration has been developed, based on the case study proposed. This allowed to highlight the challenges the paradigm entails and for the definition of suggestions to overcome them.

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Keywords: Building Information Modeling, Virtual Reality, Training, Competence, Simulation, Operator

1. INTRODUCTION

As manufacturers increase their efforts to infuse greater resilience in their operations, they are recognizing the importance of training the workforce to keep the shop floor running, especially in times of disruption (Ariansyah et al., 2024). The performance of operators in industrial settings is intricately linked to their ability to navigate and respond to the complex and unforeseen operational contexts they encounter daily. The demand for skilled operators who can efficiently and safely operate machinery, manage equipment maintenance, and adapt to ever-changing work environments is on the rise. Amidst these demands, the convergence of advanced technologies is revolutionizing training methodologies. For instance, Renault Group is going towards the first industrial Metaverse, offering stakeholders a better immersion within the working system (Renault Group, 2022). One remarkable synergy investigated in this paper is the integration of Building Information Modeling (BIM) with Virtual Reality (VR) to create BIM-enriched VR training scenes. This innovative approach leverages technology to provide immersive and highly effective training experiences. It provides a bridge between the digital and physical environments, offering a safe, faithful, and immersive platform for operators to develop their skills, make data-driven decisions, and excel in their roles. However, this approach might be time-consuming and demanding to determine the relevant information to be modeled As a result, technical challenges emerge from this integration. Motivated by the current industrial demands and the potential benefits of integrating these technologies to streamline the creation of VR training environments, this paper seeks to answer the following research question:

RQ: What are the essential aspects to consider when creating BIM-based VR environments to minimize modeling effort and maximize the effectiveness of the virtual scenes?

To answer this question, the paper will proceed as follows: Section 2 presents the theoretical background regarding VR and BIM and their current common applications. Section 3 illustrates the conceptual basis and how BIM can enrich VR models to improve the faithfulness of the scenes. Section 4 introduces the methods and materials used to develop the research and introduces the case study. Section 5 shows the preliminary results and discusses the main implications. Section 6 summarizes the findings and future works.

2. STATE OF THE ART

VR technology has become a cornerstone in training industrial workers, offering immersive environments that simulate lifelike scenarios without the associated risks of physical training environments. Studies have shown that VR training can improve trainees' confidence and engagement, particularly among novices, by offering an interactive and engaging learning experience (Radhakrishnan et al., 2021). However,

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creating a faithful, immersive environment, that keeps users focused on the scene, and effectively reflects real-world activities is a challenge for designers, and this may affect the efficacy of the training sessions. In fact, while the efficacy of VR-based training has been largely proven, implementing VR training requires overcoming technological literacy barriers among trainers and trainees, that ensure the VR content accurately reflects industrial tasks (Pedram et al., 2021), maintaining an acceptable trade-off between cost and realism is a challenge.

To solve these challenges, researchers and scientists have been implementing data-driven methods to generate, enrich and increase the faithfulness of VR environments.

More recently, BIM has emerged as a revolutionary technology and process within the architecture, engineering, and construction (AEC) industries. It represents a significant paradigm shift from traditional drafting methods to a digital representation that encompasses the structural and functional properties of infrastructure, systems, etc., thanks to its multi-dimensional feature (e.g., geometry, timetable, finance, safety, etc.) (Parsamehr et al., 2023). Traditional drafting methods, such as 3D scans or CAD, are utilized for the automatic creation of virtual replicas for digital twins (Samp.AI, n.d.), showing their high potential. However, BIM goes further the simple scope of creating the 2D/3D representation of the system of interest. It supports the life cycle management of construction projects by making it possible for users to effectively and instantly share information, allowing users to see the practical benefits of digital information management and improved communication (Rahimian et al., 2020). Among the advantages when using BIM, increased productivity, enhanced visualization, better decision-making, and communication are delivered. Recent improvements in this solution are also due to recent technical advancements like drones, photogrammetry, and laser scanning, which allow for the rapid and accurate creation of an as-built BIM representation (Alizadehsalehi et al., 2020). Advancements have explored the extension of BIM into the domain of smart built environments (SBEs), integrating IoT devices and sensors to enhance the operational efficiency and occupant comfort of buildings (Zhang, Seet, & Lie, 2015), offering new avenues for energy management, facility maintenance, and management. One of the limitations that BIM has is that even if it defines a multi-dimensional (n-D) information model, it cannot easily convey the real dimensions and proportions of the model. That's why XR technologies, and, in particular VR technologies, are used in combination with BIM, to improve users' immersion and comprehension of scale and detail (Han et al., 2021). Integrating BIM with VR has shown potential in improving training platforms for management and operations in infrastructures, particularly for emergency response scenarios such as fire evacuation in highway tunnels. By configuring BIM + VR prototypes, the training system benefits from visualization of the physical context in real time, enhancing the effectiveness of training programs (Yu et al., 2022). However, many VR applications are designed without a clear idea of how this technology may help with better learnability in terms of knowledge reusability (Ariansyah et al., 2024), and how real-world systems data can be used to improve user immersion (Johansson et al., 2024) instead of laboratory-based data. Furthermore, while a few attempts are available for training workers for management and operations in infrastructures, no solutions have been proposed yet for operations and maintenance of industrial equipment to date to the best of our knowledge.

3. ENRICHING VR SCENE WITH BIM DATA

The integration of Building Information Modeling (BIM) data into Virtual Reality (VR) training scenes is a transformative approach that holds immense potential for enhancing training effectiveness in industrial settings. The fundamental objective of a BIM model within an industrial facility is to provide a comprehensive, data-rich representation of the facility's physical attributes and operational aspects. We conceptualize the VR training scene as a digital shadow of the actual factory environment, with a profound connection between the physical and digital realms. BIM data assumes a pivotal role as a foundational source for enhancing VR training scenarios and is particularly instrumental in crafting training experiences that demand realistic spatial representations, equipment simulations, and a nuanced understanding of operational processes.

3.1 BIM Data Integration Across the VR Scene Lifecycle

Integration of BIM data into VR training scenes unfolds across key phases of the VR scene lifecycle, ensuring a holistic and dynamic training environment.

3.1.1 Template Creation Stage

In the context of VR scene development, a *template* refers to a predefined and reusable framework for creating VR training scenes. These templates serve as a starting point or blueprint for designing virtual environments. Templates can include a variety of elements such as:

1. **Spatial Layout:** The arrangement of rooms, corridors, and spaces within the virtual environment, based on the physical layout of the real-world facility.
2. **Scene composition**: This includes setting up virtual environments, placing objects, and configuring spatial relationships.
3. **Asset Placement:** The positioning of objects, equipment, machinery, and other assets within the VR scene.
4. **Interaction Design:** Designing the level of interactivity within the template, including elements such as doors, switches, controls, and interactive objects.
5. **Visual Elements:** Establishing the visual aesthetics, lighting conditions, and textures to ensure a realistic representation of the facility.

In essence, the template creation stage lays the groundwork for the VR training scene creation in accordance with the BIM data, to design a standardized structure that can later be customized and instantiated to create immersive training scenarios.

3.1.2 Instantiation Stage

The "Instantiation" stage represents a critical juncture in the creation of BIM-enriched VR training scenes. At this phase, the focus shifts towards bringing the virtual training

environment to life, ensuring that it mirrors the physical layout and operational context of the actual industrial facility as accurately as possible. Unlike the template creation stage, where the framework is established, instantiation involves populating the VR environment with the most up-to-date BIM information. Instantiation includes:

1. **Spatial Fidelity and LOD:** BIM data serves as the foundation for replicating the spatial layout of the factory or facility with a given level of detail (LOD). Every space, equipment, and structural element is imported by the BIM data, ensuring spatial fidelity, and realism.
2. **Asset Placement:** The precise location of machinery, equipment specifications, and even the orientation of objects within the VR scene are positioned based on the latest BIM data.
3. **Operational Context:** The VR scene is enriched with real time operations. This means that not only the physical assets but also the operational status of various elements, such as HVAC (Heating, Ventilation, and Air Conditioning) systems, electrical grids, and safety systems, is synchronized with the BIM data.
4. **Dynamic Updates:** As the facility undergoes changes or updates in the real world, the VR scene instantiation process can dynamically incorporate these changes through the BIM, remaining faithful to the actual facility and contributing to the continuity of training realism.

4. MATERIALS AND METHODS

The present study follows the methodological framework proposed in Fig. 1, based on the concept of digital shadow. Starting from the manufacturing site considered as a case study (described in Sec. 4.5), the BIM-based virtual replica was created and exported to a game engine, thus creating the first prototype of the system (shown in the figure on the right in Fig.1).

4.1 Creation of a BIM data model

The creation of a BIM model follows these main steps:

1. *Building the 3D Model*: the first step is to establish the basic structure of the building, in general starting from the 2D model. This involves creating floors, walls, roofs, and other essential components that form the foundation of the model.
2. *Adding Architectural Elements and Infrastructure Components*: once the basic structure is in place, additional elements and components are added. This includes doors, windows, stairs, furniture, and other architectural features. By incorporating these details, the model becomes more realistic and enables stakeholders to visualize the system more effectively.
3. *Incorporating MEP Systems*: the next step involves integrating the Mechanical, Electrical, and Plumbing (MEP) systems into the 3D model. This includes elements such as HVAC systems, electrical conduits, piping networks, and fixtures.

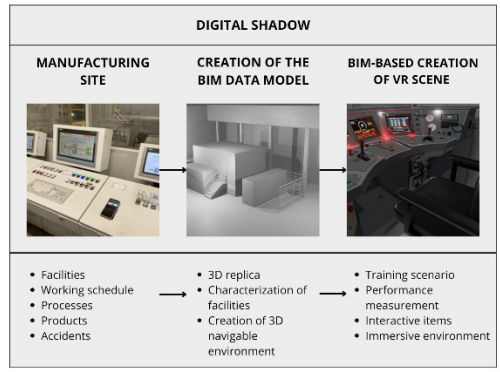


Figure 1: Methodological framework

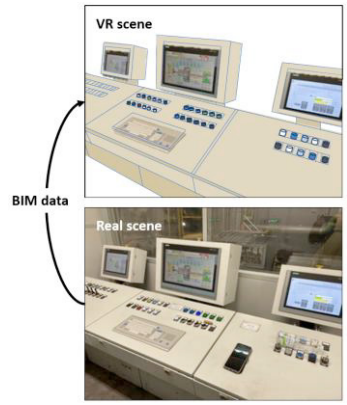


Figure 1: Cockpit of the continuous casting machinery

4. *Clash Detection and Coordination*: once all the elements are in place, clash detection is performed to identify any conflicts or collisions within the model.
5. *Integrating with other software:* BIM incorporates essential information for other software runs, such as structural data for seismic response analysis or sprinkler system details for a Fire Dynamics Simulator. Utilizing BIM as an interactive data source enables simulations that leverage geometric, structural, and MEP data for comprehensive analyses.

With respect to the defined macro process, the data that could be extrapolated from BIM in order to activate the simulation via VR are shown in Table 1.

4.2 BIM-based creation of VR scene templates

To configure virtual training scenes automatically, we developed pre-configured scene templates for each game level. We propose here a complete workflow for the creation of VR scene templates to be enriched with BIM data:

1. **Design the Template Scene with BIM-based Spatial Layout Data:** based on the given 3D BIM representations, the initial template is initialized.
2. **Organize Assets with BIM-based Placement Data and Metadata:** 3D models, textures, and other assets within the project hierarchy are initialized. This includes the asset placement and the asset metadata (e.g., equipment specifications, maintenance requirements, and operational details).
3. **Configure Prefabs with BIM-based Asset Properties:** Objects and elements are converted into

Prefabs, storing specific settings and behaviors, specified within the BIM, for easy duplication across scenes.

4. **Set Up Interactions using BIM-based interaction data:** This includes scripting actions like opening doors, activating machinery, or simulating equipment behavior, based on the BIM data available.
5. **Define Lighting and Ambiance with BIM-based Ambient Data:** This data can be used to replicate realistic lighting conditions and ambiance in the template scene, based on the features of assets defined within the BIM.
6. **Design User Interface (UI) with BIM-based control panels and labels:** BIM can provide information about control panels, buttons, and labels within the facility. This data can guide the design of UI elements and instructional pop-ups in the VR template (see Figure 1).

Once the template scene is saved, it can be duplicated and customized with other assets, interactions, or environmental elements as needed, always ensuring that the core structure and assets remain consistent.

4.3 Optimizing BIM data for VR training

Optimizing BIM data for VR training involves making choices regarding the nature of the data (static or dynamic) and the LOD to match the specific requirements of the training scenarios. A high LOD includes high visual fidelity (texture, lighting, shadows, material properties), environmental realism (atmospheric agents video and sound), and more. Instead, a *sufficient LOD* refers to basic visual fidelity (not over-detailed resolution textures, no complex shadowing or lightning features, simple material rendering), functional accuracy (focus primarily on necessary interactive elements to perform the task), and more. The goal of the present application is the training of an industrial non-repetitive process. In this case, the LOD necessary is not high, but sufficient so that the operator can complete the different activities, recognize the different physical commands, evaluate the different indicators, recognize his/her position in the workspace, etc.

A sufficient LOD provides the necessary fidelity for trainees to perform tasks accurately but without overwhelming system resources (Graham et al., 2018). Johansson and Roupé (2024) define three technical challenges/barrier areas:

- Interaction (e.g. user interface, tools): it defines the common features among BIM and VR-platform such as teleport navigation, miniature models for overview and navigation, the tools palette, etc.;
- Interactivity (e.g. frame rate, rendering performance): it defines the sufficient frame rate to have a high-quality experience.
- Interoperability (e.g. BIM-to-VR conversion, data transfer, metadata): it defines the correctness and completeness of the data transfer process.

Second, in some VR training scenarios, the information from the BIM model may not change over time or may not need to be updated in real time. This situation occurs when the BIM data used for VR training is related to aspects of the environment that remain constant during training sessions.

Table 1: Potential extrapolated data from BIM

Table 2: Potential extrapolated data from BIM

Class of data extrapolated		Specific data type
Spatial data	**Spatial attributes**: to evaluate the operators' position in the plant layout	• 3D room layout • Door and window locations and sizes • Lighting fixture types and location • Data cabling infrastructure
	Furnishings and equipment: to recognize the position of the different cockpit commands and reach them	• Types and dimensions of control desks • Number and types of control panels, including dimensions and manufacturers • Types and sizes of computer monitors • Specifications for any control hardware (buttons, switches, levers) • Details on any communication devices
	Operational details: to recognize visual alarms	• Information on the processes controlled by the room • Functional specifications for the control systems • Software systems used for monitoring and control • Emergency power systems
Auxiliary and production systems data	**Specific limits of the production systems**	• Max/min velocity of the continuous casting • Max/min temperature of the ovens
	Specific limits of the auxiliary systems	• Max pressures of the refrigerant fluid • Max velocity of the refrigerant fluid
	Environmental control	• HVAC systems • Air quality sensors or controls
	Safety system	• Fire suppression systems (e.g., sprinklers, extinguishers) • Emergency egress routes • Safety signage
	Maintenance system	• Probability of breakdowns of the different systems (auxiliary and production) • Service history of equipment • Schedules for maintenance checks • Details of equipment warranties and service contracts

For example, if BIM data is used to define the 3D layout of a factory floor, this layout may not change frequently. In this case, BIM data can be exported as a static representation. This simplifies the integration process and ensures that the VR scene reflects a snapshot of the BIM data at the time of export. On the other hand, there are scenarios where the BIM model itself changes over time or needs to be updated in real time to accurately represent the facility. This is especially relevant when simulating dynamic systems, MEP (Mechanical, Electrical, and Plumbing) data, or processes that evolve. For instance, if a manufacturing process is simulated, where the status of machines and equipment changes, or if operators are trained on HVAC systems where temperature and airflow data fluctuate, *dynamic BIM data* are needed. In such cases, the VR scene must be implemented to fetch real time updates from the BIM model. This typically involves the creation of a dedicated

script enabling the VR environment to receive updated data from the BIM model as changes occur. These integrations can be done through APIs, databases, etc.

4.4 BIM-to-VR integration

After defining the main features that the models must have to reach the goal of this project, the first step is to choose the right software for BIM and VR implementation. To choose among the several platforms available, four criteria were used.

- Cost: considering the budget available for the project, the cost of software has to be considered.
- Interoperability: capability of the platforms to exchange information consistently among them.
- Usability: easiness and intuitiveness of the software.
- Reliability: related to the stability, compatibility with the operating systems, and relevance of the applications.

Based on the selection criteria, Blender and its game engine (UPBGE), as well as Unity, together with Autodesk Revit are the best tools for the purpose of the project.

The second step is to find a way to integrate these two software and the related model generated. Two main methods are available: file-based, using the .IFC extension, or addin-based, where plugins are used to directly start a VR session from the BIM platform (Alizadehsalehi et al., 2020). The first method is more common when dealing directly with game engines (e.g., Blender UPBGE, Unity, Unreal Engine), while the second method is more used for direct-to-VR solutions. When dealing with the file-based method, the import speed has to be considered, since importing a large classic 3D file (e.g., FBX) (~90,000,000 triangles) from Revit can last around 4 hours (Zou et al., 2018). Therefore, this method can be more efficient when dealing with smaller models, as is the use case treated in this study. In addition, special attention should be paid to data transmission. For example, the inclusion of LED lamps within the BIM file is not associated with elements with lighting features within the player. This means that, in principle, there is no automatic association between 3D elements or information and function.

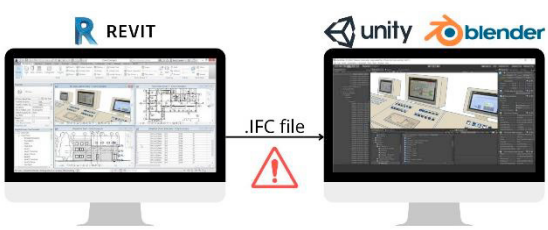


Figure 2: Integration BIM-VR

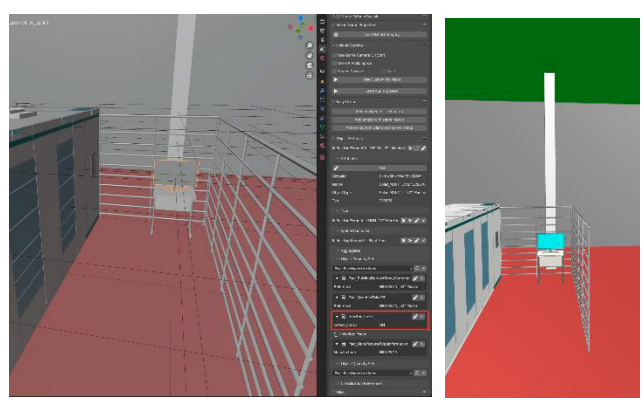
Another output is related to the availability of plugins that can directly integrate BIM data into VR platforms. To the best of our knowledge, we noticed a clear lack of easy-to-use and immediate solutions for the problem, thus detecting a crucial gap in the state of the art.

4.5 Case study

This research focuses on the project entitled "pREScriptive dIgitaL twins for cognItive-enriched compEtency developmeNt of workforCe of the future in smart factoriEs" (RESILIENCE). It delves into the integration of prescriptive analytics, digital twin (DT) technologies, building information modeling (BIM), and on-the-job training (OJT) for nonrecurrent and critical tasks. A case study, conducted in an Italian manufacturing company specializing in drawn parts and brass extrusion will be used for developing the first prototype of the system. The case analyzed involves the casting process and its malfunctioning, in which humans might contribute to unavailability and system damage. The VR simulator will specifically address the recovery activities on the oven. The learning process involves recognizing anomalies in the mechanical driver by monitoring continuous casting cockpit performance and physically inspecting the driver's movement. The methodology proposed in the sections above is applied to firstly, develop the BIM of the manufacturing site, and, secondly, to export it into the game engine to characterize the training environment.

5. PRELIMINARY RESULTS AND DISCUSSION

Addressing the OJT paradigm proposed within the RESILIENCE project, our approach streamlines the creation of customizable BIM-enriched VR scenes. Following the methodology described in the previous sections, the prototype of a BIM-based VR scene was created. The representation of the manufacturing plant was created in Revit and exported in Blender as an IFC file. This allowed the preservation of all the additional information related to the site assets. For instance, Fig. 4a shows how the object (monitor) status is set to "on", and, therefore, in the player mode (see Fig. 4b) it is turned on. The same principle can be applied to all the other objects to characterize their status for the VR training session. This can support the need for real time data updates from the real system, or for setting different levels of difficulty during the training (e.g., additional action required by the user to turn on the monitor).



(a) (b)

Figure 4: Object status defined in BIM (a) visualized in the player mode (b)

From the development of the prototype, the following aspects can be derived:

1. The existence of a CAD (e.g. layout) or a BIM model (defined for other objectives) of the manufacturing site can be considered a prerequisite for reducing the effort of modeling BIM-enriched VR training environments;
2. When dealing with BIM, .IFC extension has to be used to preserve the additional information related to

the assets. Indeed, using FBX files implies that only geometrical features will be exported.

3. Some data in the BIM model might not be necessary for the training environment. This depends on the specific case study under analysis and the LOD chosen.
4. A middleware is shown to be necessary to automatically "translate" information or objects available in the BIM into relevant assets in the game engine (e.g. lamps into light objects for the game engine with the related lightning function).

6. CONCLUSIONS

Integrating BIM with VR presents a promising technology for training in complex and non-repetitive operations. Current training solutions, such as the transmission of knowledge from highly-skilled to less-skilled employees, particularly for complex and hazardous tasks, often result in inefficient skill retention and an increased likelihood of mistakes. Providing safe hands-on solutions, even for intricate tasks, offers a more effective learning method. Why are BIM-enriched VR training scenes important? Using a realistic model, as opposed to traditional 3D models, can significantly improve immersion and familiarity with the environment, thereby enhancing skill retention. While traditional integration of BIM and VR is common in co-design for complex buildings, our approach pioneers the use of BIM-Enriched VR Scenes for training non-repetitive operations in industrial settings. This enables tailored training for manufacturing processes, equipment operation, and emergency response procedures, aligning with the unique characteristics of each facility. The proposed method's novelty and adaptability mark a scientific contribution to ease the creation of effective industrial training. Moving forward, implementing and evaluating this approach in a real industrial case will illuminate its practical pros and cons, paving the way for further advancements in easing the generation of automatic training scenes.

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